

Diag Solutions

1. **1.** Two poles of height 6 m and 11 m stand on a plane ground. If the distance between their feet is 12 m, find the distance between their tops.

Answer: (B)

Solution:

1. Let the poles be $AB = 6$ m and $CD = 11$ m. The distance between them is $BD = 12$ m.
2. Draw a line from A parallel to BD meeting CD at E .
3. Then $CE = CD - ED = 11 - 6 = 5$ m.
4. In right $\triangle AEC$, $AC^2 = AE^2 + CE^2 = 12^2 + 5^2 = 144 + 25 = 169$.
5. $AC = 13$ m.

Derivation:

$$AC = \sqrt{12^2 + (11 - 6)^2} = \sqrt{144 + 25} = \sqrt{169} = 13$$

Triangles — Pythagoras Theorem

2. **2.** In $\triangle ABC$, D and E are points on sides AB and AC respectively such that $AD = 3$ cm, $DB = 2$ cm, $AE = 4.5$ cm, and $EC = 3$ cm. Which of the following is true?

Answer: (A)

Solution:

1. Check if $\frac{AD}{DB} = \frac{AE}{EC}$.
2. $\frac{AD}{DB} = \frac{3}{2} = 1.5$.
3. $\frac{AE}{EC} = \frac{4.5}{3} = 1.5$.
4. Since the ratios are equal, $DE \parallel BC$ by the Converse of BPT.

Derivation:

$$\frac{AD}{DB} = \frac{3}{2} = 1.5 \text{ and } \frac{AE}{EC} = \frac{4.5}{3} = 1.5 \\ \Rightarrow DE \parallel BC$$

Triangles — Converse of BPT

3. **3.** If $\triangle ABC \sim \triangle DEF$, $\angle A = 50^\circ$, and $\angle E = 70^\circ$, then $\angle C$ is:

Answer: (C)

Solution:

1. Since $\triangle ABC \sim \triangle DEF$, corresponding angles are equal: $\angle A = \angle D$, $\angle B = \angle E$, $\angle C = \angle F$.

2. Given $\angle E = 70^\circ$, so $\angle B = 70^\circ$.
3. In $\triangle ABC$, $\angle A + \angle B + \angle C = 180^\circ$.
4. $50^\circ + 70^\circ + \angle C = 180^\circ$.
5. $\angle C = 180^\circ - 120^\circ = 60^\circ$.

Derivation:

$$\angle C = 180^\circ - (\angle A + \angle B) = 180^\circ - (50^\circ + 70^\circ) = 60^\circ$$

Triangles — Similarity of Triangles

4. **4.** Assertion (A): If triangle $ABC \sim$ triangle DEF and $AB = 2$ cm, $DE = 4$ cm, then the ratio of their areas is $1 : 4$. Reason (R): The ratio of the areas of two similar triangles is equal to the square of the ratio of their corresponding sides.

Answer: (A)

Solution:

1. State the theorem regarding the ratio of areas of similar triangles.
2. Calculate the square of the ratio of sides.
3. Compare with the given ratio of areas.

Derivation: Ratio of areas $= (AB/DE)^2 = (2/4)^2 = (1/2)^2 = 1/4 = 1:4$ \\ Since both Assertion and Reason are true and Reason explains the Assertion, option (A) is correct.

Triangles — Areas of Similar Triangles

5. **5.** Assertion (A): In triangle ABC , if $DE \parallel BC$ with D on AB and E on AC , then $AD/DB = AE/EC$. Reason (R): Thales Theorem (Basic Proportionality Theorem) states that if a line is drawn parallel to one side of a triangle intersecting the other two sides, then it divides the two sides in the same ratio.



Answer: (A)

Solution:

1. Identify the theorem being used.
2. Verify the statement of the Thales Theorem.
3. Check if the Assertion is the direct application of the Theorem.

Derivation: Assertion is the mathematical statement of the Basic Proportionality Theorem. $\backslash\backslash$ Reason is the definition of the Basic Proportionality Theorem. $\backslash\backslash$ Both are true and R is the correct explanation. *Triangles — Basic Proportionality Theorem*

6. **6.** Assertion (A): Two triangles are similar if their corresponding angles are equal. Reason (R): If the corresponding sides of two triangles are in the same ratio, then the triangles are congruent.

Answer: (C)

Solution:

1. Check the validity of the AAA similarity criterion.
2. Check the validity of the SSS similarity vs congruence distinction.
3. Determine the truth value of both statements.

Derivation: Assertion is true by AAA similarity criterion. $\backslash\backslash$ Reason is false because proportional sides imply similarity, not necessarily congruence. $\backslash\backslash$ Therefore, A is true but R is false. *Triangles — Criteria for Similarity of Triangles*

7. **7.** Assertion (A): In triangle ABC, if $\angle A = 90^\circ$ and $AD \perp BC$, then triangle ADB $\sim$ triangle CDA. Reason (R): Two triangles are similar if two angles of one triangle are respectively equal to two angles of the other triangle.

Answer: (A)

Solution:

1. Apply the AA similarity criterion for right triangles.
2. Verify the angle relationships.
3. Confirm the reasoning.

Derivation: In triangle ADB and triangle CDA, $\angle ADB = \angle CDA = 90^\circ$. $\angle BAD = \angle ACD$ (both complement $\angle DAC$). $\backslash\backslash$ By AA similarity, triangle ADB $\sim$ triangle CDA. $\backslash\backslash$ Reason is the AA criterion definition. *Triangles — Similarity of Triangles*

8. **8.** Assertion (A): The perimeters of two similar triangles are in the same ratio as their corresponding sides. Reason (R): If triangle ABC $\sim$ triangle PQR, then $AB/PQ = BC/QR = CA/RP = (AB + BC + CA)/(PQ + QR + RP)$.

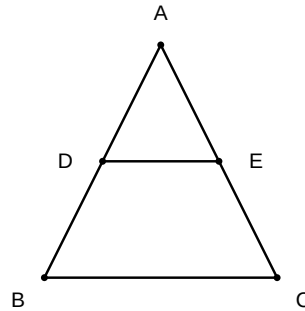
Answer: (A)

Solution:

1. Use the properties of ratios.
2. Verify the perimeter formula.
3. Check the relationship between the assertion and the reason.

Derivation: Given triangle ABC $\sim$ triangle PQR, $\backslash\backslash$ $AB/PQ = BC/QR = CA/RP = k$. $\backslash\backslash$ $AB=kPQ$, $BC=kQR$, $CA=kRP$. $\backslash\backslash$ $\text{Perimeter(ABC)} = AB+BC+CA = k(PQ+QR+RP) = k \text{ times Perimeter(PQR)}$. $\backslash\backslash$ Ratio = k. $\backslash\backslash$ Both are true and R explains A. *Triangles — Similarity of Triangles*

9. 9. In $\triangle ABC$, $DE \parallel BC$ such that $AD = 3 \text{ cm}$, $DB = 4 \text{ cm}$, and $AE = 6 \text{ cm}$. Find the length of EC .



Answer:

$$8\text{cm}$$

Solution:

1. Apply the Basic Proportionality Theorem (BPT) since $DE \parallel BC$.
2. Substitute the given values into $\frac{AD}{DB} = \frac{AE}{EC}$ and solve for EC .

Derivation:

$$\begin{aligned}\frac{AD}{DB} &= \frac{AE}{EC} \\ \Rightarrow \frac{3}{4} &= \frac{6}{EC} \\ \Rightarrow 3 \times EC &= 24 \\ \Rightarrow EC &= 8\text{cm}\end{aligned}$$

Triangles — Basic Proportionality Theorem

10. 10. Two triangles $\triangle ABC$ and $\triangle DEF$ are similar. If $\text{Area}(\triangle ABC) = 64 \text{ cm}^2$ and $\text{Area}(\triangle DEF) = 121 \text{ cm}^2$, and $EF = 11 \text{ cm}$, find the length of BC .

Answer:

$$8\text{cm}$$

Solution:

1. Use the theorem: The ratio of areas of two similar triangles is equal to the square of the ratio of their corresponding sides.
2. Set up the equation $\frac{\text{Area}(\triangle ABC)}{\text{Area}(\triangle DEF)} = \left(\frac{BC}{EF}\right)^2$ and solve for BC .

Derivation:

$$\begin{aligned}
\frac{64}{121} &= \left(\frac{BC}{11}\right)^2 \\
\Rightarrow \sqrt{\frac{64}{121}} &= \frac{BC}{11} \\
\Rightarrow \frac{8}{11} &= \frac{BC}{11} \\
\Rightarrow BC &= 8cm
\end{aligned}$$

Triangles — Areas of Similar Triangles

11. 11. In $\triangle PQR$, S and T are points on sides PQ and PR respectively such that $\angle PST = \angle PQR$. If $PS = 2\text{ cm}$, $SQ = 3\text{ cm}$, and $PT = 1.5\text{ cm}$, find the length of TR .

Answer:

$$2.25cm$$

Solution:

1. Prove $\triangle PST \sim \triangle PQR$ by AA similarity criterion since $\angle P = \angle P$ and $\angle PST = \angle PQR$.
2. Use the proportionality ratio $\frac{PS}{PQ} = \frac{PT}{PR}$ where $PQ = PS + SQ$ and $PR = PT + TR$.

Derivation:

$$\begin{aligned}
\frac{2}{2+3} &= \frac{1.5}{1.5+TR} \\
\Rightarrow \frac{2}{5} &= \frac{1.5}{1.5+TR} \\
\Rightarrow 3+2TR &= 7.5 \\
\Rightarrow 2TR &= 4.5 \\
\Rightarrow TR &= 2.25cm
\end{aligned}$$

Triangles — Criteria for Similarity of Triangles

12. 12. A vertical pole of length 6 m casts a shadow 4 m long on the ground and at the same time a tower casts a shadow 28 m long. Find the height of the tower.

Answer:

$$42m$$

Solution:

1. Recognize that the triangles formed by the objects and their shadows are similar due to the same sun angle.
2. Set up the proportion $\frac{\text{Height}_1}{\text{Shadow}_1} = \frac{\text{Height}_2}{\text{Shadow}_2}$ and solve for the tower height.

Derivation:

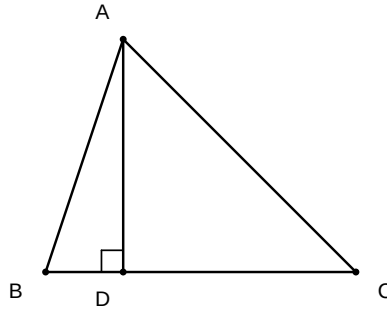
$$\frac{6}{4} = \frac{h}{28}$$

$$\Rightarrow 4h = 168$$

$$\Rightarrow h = 42m$$

Triangles — Applications of Similar Triangles

13. 13. In $\triangle ABC$, $AD \perp BC$. Prove that $AB^2 + CD^2 = AC^2 + BD^2$.



Answer:

Proof based on Pythagoras Theorem

Solution:

1. Apply Pythagoras theorem in $\triangle ABD$ ($AB^2 = AD^2 + BD^2$) and $\triangle ACD$ ($AC^2 = AD^2 + CD^2$).
2. Express AD^2 from both equations and equate them to show the result.

Derivation:

$$AB^2 - BD^2 = AD^2$$

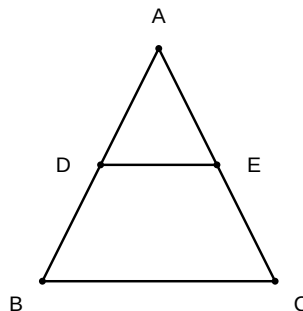
$$AC^2 - CD^2 = AD^2$$

$$AB^2 - BD^2 = AC^2 - CD^2$$

$$AB^2 + CD^2 = AC^2 + BD^2$$

Triangles — Pythagoras Theorem

14. 14. In $\triangle ABC$, $DE \parallel BC$ such that $AD = (4x - 3)$ cm, $AE = (8x - 7)$ cm, $BD = (3x - 1)$ cm, and $CE = (5x - 3)$ cm. Find the value of x .



Answer:

$$x = 1 \text{ or } x = 0.8$$

Solution:

1. Apply Thales Theorem (Basic Proportionality Theorem) since $DE \parallel BC$.
2. Set up the ratio $\frac{AD}{BD} = \frac{AE}{CE}$.
3. Solve the resulting quadratic equation for x .

Derivation:

$$\begin{aligned}\frac{4x-3}{3x-1} &= \frac{8x-7}{5x-3} \\ (4x-3)(5x-3) &= (8x-7)(3x-1) \\ 20x^2 - 12x - 15x + 9 &= 24x^2 - 8x - 21x + 7 \\ 20x^2 - 27x + 9 &= 24x^2 - 29x + 7 \\ 4x^2 - 2x - 2 &= 0 \\ 2x^2 - x - 1 &= 0 \\ (2x+1)(x-1) &= 0 \\ x = 1, x = -0.5\end{aligned}$$

Triangles — Basic Proportionality Theorem

15. 15. Two poles of height 6 m and 11 m stand on a plane ground. If the distance between their feet is 12 m, find the distance between their tops.

Answer:

$$13 \text{ m}$$

Solution:

1. Represent the poles as vertical segments on a horizontal ground.
2. Form a right-angled triangle by drawing a line parallel to the ground from the top of the shorter pole.
3. Use the Pythagorean theorem to calculate the hypotenuse.

Derivation:

$$\begin{aligned}\text{Let } AB &= 6 \text{ m and } CD = 11 \text{ m be the poles.} \\ \text{Distance between feet } AC &= 12 \text{ m.} \\ \text{Draw } BE \perp CD \text{ such that } CE &= AB = 6 \text{ m.} \\ DE &= CD - CE = 11 - 6 = 5 \text{ m.} \\ \text{In right } \triangle BED, BD^2 &= BE^2 + DE^2 \\ BD^2 &= 12^2 + 5^2 = 144 + 25 = 169 \\ BD &= \sqrt{169} = 13 \text{ m}\end{aligned}$$

Triangles — Pythagoras Theorem

16. **16.** In $\triangle ABC$, $\angle A = 90^\circ$ and $AD \perp BC$. If $AD = 8$ cm and $BD = 4$ cm, find the length of CD .

Answer:

16 cm

Solution:

1. Use the property of similarity in a right triangle where the altitude to the hypotenuse divides the triangle into two triangles similar to the whole and each other.
2. Establish the ratio $\triangle ADB \sim \triangle CDA$.
3. Solve for CD using $\frac{AD}{BD} = \frac{CD}{AD}$.

Derivation:

In $\triangle ABC$, $\angle A = 90^\circ$ and $AD \perp BC$.

$$\triangle ADB \sim \triangle CDA$$

$$\frac{AD}{CD} = \frac{BD}{AD}$$

$$AD^2 = BD \times CD$$

$$8^2 = 4 \times CD$$

$$64 = 4 \times CD$$

$$CD = 16 \text{ cm}$$

Triangles — Similarity of Triangles

17. **17.** The perimeters of two similar triangles are 30 cm and 20 cm respectively. If one side of the first triangle is 9 cm, find the length of the corresponding side of the second triangle.

Answer:

6 cm

Solution:

1. Recall that the ratio of perimeters of two similar triangles is equal to the ratio of their corresponding sides.
2. Set up the proportion $\frac{P_1}{P_2} = \frac{S_1}{S_2}$.
3. Substitute the given values and solve for S_2 .

Derivation:

$$\frac{P_1}{P_2} = \frac{S_1}{S_2}$$

$$\frac{30}{20} = \frac{9}{S_2}$$

$$30 \times S_2 = 20 \times 9$$

$$30 \times S_2 = 180$$

$$S_2 = \frac{180}{30} = 6 \text{ cm}$$

18. 18. In $\triangle ABC$, P and Q are points on sides AB and AC such that $PQ \parallel BC$. If $AP = 3$ cm, $PB = 2$ cm and $AQ = 2.7$ cm, find AC .

Q
•

Answer:

4.5 cm

Solution:

1. Apply the Thales theorem: $\frac{AP}{AB} = \frac{AQ}{AC}$.
2. Calculate $AB = AP + PB = 3 + 2 = 5$ cm.
3. Solve the ratio for AC .

Derivation:

$$\begin{aligned}\frac{AP}{AP + PB} &= \frac{AQ}{AC} \\ \frac{3}{3 + 2} &= \frac{2.7}{AC} \\ \frac{3}{5} &= \frac{2.7}{AC} \\ 3 \times AC &= 5 \times 2.7 \\ 3 \times AC &= 13.5 \\ AC &= 4.5 \text{ cm}\end{aligned}$$

19. 19. Prove that if a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, the other two sides are divided in the same ratio. Use this theorem to find the length of EC in triangle ABC where $DE \parallel BC$, $AD = 3$ cm, $DB = 4$ cm, and $AE = 6$ cm.



Answer:

$$EC = 8cm$$

Solution:

1. State Basic Proportionality Theorem (BPT).
2. Construct the proof by drawing altitudes and using area ratios.
3. Apply the theorem to the given values: $\frac{AD}{DB} = \frac{AE}{EC}$.
4. Substitute the given values: $\frac{3}{4} = \frac{6}{EC}$.
5. Solve for EC .

Derivation:

$$\begin{aligned}\frac{AD}{DB} &= \frac{AE}{EC} \\ \frac{3}{4} &= \frac{6}{EC} \\ 3 \times EC &= 6 \times 4 \\ 3 \times EC &= 24 \\ EC &= \frac{24}{3} \\ EC &= 8cm\end{aligned}$$

Triangles — Basic Proportionality Theorem

- 20. 20.** In $\triangle ABC$, $\angle A = 90^\circ$ and $AD \perp BC$. Prove that $AD^2 = BD \times CD$. If $BD = 4$ cm and $CD = 9$ cm, calculate the length of AD .

Answer:

$$AD = 6cm$$

Solution:

1. Use the property of similar triangles: $\triangle ADB \sim \triangle CDA$.
2. Write the ratio of corresponding sides: $\frac{AD}{CD} = \frac{BD}{AD}$.
3. Cross multiply to get $AD^2 = BD \times CD$.

4. Substitute the given values $BD = 4$ and $CD = 9$.
5. Calculate $AD = \sqrt{36} = 6$ cm.

Derivation:

$$\begin{aligned}\triangle ADB &\sim \triangle CDA \\ \frac{AD}{CD} &= \frac{BD}{AD} \\ AD^2 &= BD \times CD \\ AD^2 &= 4 \times 9 \\ AD^2 &= 36 \\ AD &= 6\text{cm}\end{aligned}$$

Triangles — Similarity of Triangles

- 21. 21.** Two poles of heights a meters and b meters are p meters apart. Prove that the height of the point of intersection of the lines joining the top of each pole to the foot of the opposite pole is given by $\frac{ab}{a+b}$ meters.

Answer:

$$\text{Height} = \frac{ab}{a+b}$$

Solution:

1. Let the poles be $AB = a$ and $CD = b$. Let them be p apart.
2. Let $EF = h$ be the height of the intersection point.
3. Show $\triangle EFC \sim \triangle ABC$ and $\triangle EFD \sim \triangle ABD$.
4. Express $\frac{EF}{AB} = \frac{FC}{BC}$ and $\frac{EF}{CD} = \frac{FD}{BD}$.
5. Sum the ratios to eliminate the segments of p and solve for h .

Derivation:

$$\begin{aligned}\frac{h}{a} &= \frac{FC}{p} \text{ and } \frac{h}{b} = \frac{FD}{p} \\ \frac{h}{a} + \frac{h}{b} &= \frac{FC + FD}{p} \\ h\left(\frac{1}{a} + \frac{1}{b}\right) &= \frac{p}{p} = 1 \\ h\left(\frac{a+b}{ab}\right) &= 1 \\ h &= \frac{ab}{a+b}\end{aligned}$$

Triangles — Similarity and Ratios

- 22. 22.** Prove that the ratio of the areas of two similar triangles is equal to the square of the ratio of their corresponding sides. Using this, find the ratio of the areas of two similar triangles if their sides are in the ratio 4 : 9.

Answer:

$$\text{Ratio} = 16 : 81$$

Solution:

1. State the theorem for the ratio of areas of similar triangles.
2. Draw two similar triangles and their altitudes.
3. Use similarity to show $\frac{\text{Area}_1}{\text{Area}_2} = \left(\frac{\text{side}_1}{\text{side}_2}\right)^2$.
4. Apply the given ratio $\frac{s_1}{s_2} = \frac{4}{9}$.
5. Calculate the square of the ratio: $\left(\frac{4}{9}\right)^2 = \frac{16}{81}$.

Derivation:

$$\begin{aligned}\frac{ar(\triangle ABC)}{ar(\triangle PQR)} &= \left(\frac{AB}{PQ}\right)^2 \\ \frac{ar(\triangle ABC)}{ar(\triangle PQR)} &= \left(\frac{4}{9}\right)^2 \\ \frac{ar(\triangle ABC)}{ar(\triangle PQR)} &= \frac{16}{81}\end{aligned}$$

Triangles — Areas of Similar Triangles

23. 23. The discriminant of the quadratic equation $3x^2 - 4x + 1 = 0$ is

Answer: (C)

Solution:

1. Identify the coefficients a , b , and c from the standard form $ax^2 + bx + c = 0$.
2. The coefficients are $a = 3$, $b = -4$, and $c = 1$.
3. Use the formula for the discriminant $D = b^2 - 4ac$.
4. Substitute the values into the formula: $D = (-4)^2 - 4(3)(1)$.
5. Calculate the result: $16 - 12 = 4$.

Derivation:

$$\begin{aligned}a &= 3, b = -4, c = 1 \\ D &= b^2 - 4ac \\ D &= (-4)^2 - 4(3)(1) \\ D &= 16 - 12 \\ D &= 4\end{aligned}$$

Quadratic Equations — Nature of Roots

24. 24. The discriminant of the quadratic equation $3x^2 - 5x + 2 = 0$ is

Answer: (B)

Solution:

1. Identify the coefficients a , b , and c from the standard form $ax^2 + bx + c = 0$.

2. The coefficients are $a = 3$, $b = -5$, and $c = 2$.
3. Use the formula for the discriminant $D = b^2 - 4ac$.
4. Substitute the values into the formula: $D = (-5)^2 - 4(3)(2)$.
5. Calculate the result: $D = 25 - 24 = 1$.

Derivation:

$$\begin{aligned}a &= 3, b = -5, c = 2 \\D &= b^2 - 4ac \\D &= (-5)^2 - 4(3)(2) \\D &= 25 - 24 \\D &= 1\end{aligned}$$

Quadratic Equations — Nature of Roots

- 25. 25.** The discriminant of the quadratic equation $2x^2 - 5x + 3 = 0$ is

Answer: (A)

Solution:

1. Identify the coefficients a , b , and c from the standard form $ax^2 + bx + c = 0$.
2. Apply the formula for the discriminant $D = b^2 - 4ac$.
3. Substitute the values $a = 2$, $b = -5$, and $c = 3$ into the formula.
4. Calculate the final value.

Derivation:

$$\begin{aligned}a &= 2, b = -5, c = 3 \\D &= b^2 - 4ac \\D &= (-5)^2 - 4 \times 2 \times 3 \\D &= 25 - 24 \\D &= 1\end{aligned}$$

Quadratic Equations — Nature of Roots

- 26. 26.** If the prime factorization of a positive integer n is given by $n = 2^3 \times 3^2 \times 5^k$ and it is known that n is a multiple of 150, then the smallest possible value of the non-negative integer k is:

Answer: (C)

Solution:

1. Find the prime factorization of 150.
2. Express n in terms of its prime factors.
3. Use the definition of a multiple to establish the condition for k .
4. Compare the exponents of the primes to find the smallest value of k .

Derivation: $150 = 15 \times 10 = (3 \times 5) \times (2 \times 5) = 2^1 \times 3^1 \times 5^2$ \ Given $n = 2^3 \times 3^2 \times 5^k$ \ For n to be a multiple of 150, the exponent of each prime factor $3 \geq 1$ (True) For factor 3: $2 \geq 1$ (True) For factor 5: $k \geq 2$ The smallest integer k satisfying $k \geq 2$ is 2. *Real Numbers — Fundamental Theorem of Arithmetic*

- 27. 27.** The roots of the quadratic equation $x^2 - 7x + 12 = 0$ are obtained by factorising the middle term. The roots of the equation are:

Answer: (B)

Solution:

1. Identify the quadratic equation: $x^2 - 7x + 12 = 0$.
2. Find two numbers whose product is 12 and whose sum is -7 . These numbers are -3 and -4 .
3. Rewrite the middle term: $x^2 - 3x - 4x + 12 = 0$.
4. Factor by grouping: $x(x - 3) - 4(x - 3) = 0$.
5. Factor out the common binomial: $(x - 3)(x - 4) = 0$.
6. Solve for x : $x - 3 = 0$ or $x - 4 = 0$, which gives $x = 3$ or $x = 4$.

Derivation:

$$\begin{aligned} x^2 - 7x + 12 &= 0 \\ x^2 - 3x - 4x + 12 &= 0 \\ x(x - 3) - 4(x - 3) &= 0 \\ (x - 3)(x - 4) &= 0 \\ x &= 3, 4 \end{aligned}$$

Quadratic Equations — factorisation method

- 28. 28.** Assertion (A): For any two positive integers a and b , if $HCF(a, b) = 11$ and $LCM(a, b) = 693$, then the product $a \times b = 7623$. Reason (R): For any two positive integers a and b , $HCF(a, b) \times LCM(a, b) = a \times b$.

Answer: (A)

Solution:

1. Identify the relationship between HCF, LCM and the product of two numbers.
2. Check the assertion: $11 \times 693 = 7623$.
3. Verify the Reason: The property $HCF(a, b) \times LCM(a, b) = a \times b$ is a standard theorem for positive integers.
4. Conclude that both are true and the reason explains the assertion.

Derivation:

$$11 \times 693 = 7623$$

Since $HCF(a, b) \times LCM(a, b) = a \times b$ is always true, the assertion follows from the reason.

Real Numbers — Fundamental Theorem of Arithmetic

29. 29. Show that for any positive integer n , $n^3 - n$ is divisible by 6.

Answer:

$n^3 - n = (n - 1)n(n + 1)$, which is a product of three consecutive integers, hence divisible by $3! = 6$.

Solution:

1. Factorize the expression $n^3 - n$ as $n(n^2 - 1)$.
2. Further factorize to get $n(n - 1)(n + 1)$.
3. Identify that this is the product of three consecutive integers.
4. State that in any set of three consecutive integers, at least one must be even and exactly one must be a multiple of three.
5. Conclude that the product must be divisible by $2 \times 3 = 6$.

Derivation:

$$\begin{aligned}n^3 - n &= n(n^2 - 1) \\ &= (n - 1)n(n + 1)\end{aligned}$$

Since $(n - 1), n, (n + 1)$ are three consecutive integers, their product is divisible by $3! = 6$.

Real Numbers — Euclid's Division Lemma

30. 30. Find the greatest number of 6 digits which is exactly divisible by 12, 15, and 20.

Answer:

999960

Solution:

1. Find the LCM of 12, 15, and 20.
2. $12 = 2^2 \times 3$, $15 = 3 \times 5$, $20 = 2^2 \times 5$.
3. $LCM = 2^2 \times 3 \times 5 = 60$.
4. Identify the largest 6-digit number as 999999.
5. Divide 999999 by 60 and find the remainder.
6. Subtract the remainder from 999999 to get the largest divisible number.

Derivation:

$$\begin{aligned}LCM(12, 15, 20) &= 60 \\ 999999 \div 60 &= 16666 \text{ with remainder } 39 \\ 999999 - 39 &= 999960\end{aligned}$$

Real Numbers — HCF and LCM

- 31. 31.** A merchant has 120 litres of oil of one kind, 180 litres of another kind and 240 litres of a third kind. He wants to sell the oil by filling the three kinds of oil in tins of equal capacity. What should be the greatest capacity of such a tin?

Answer:

60litres

Solution:

1. Recognize that the problem requires finding the greatest common measure.
2. Identify that this is the HCF of 120, 180, and 240.
3. Prime factorize the numbers: $120 = 2^3 \times 3 \times 5$, $180 = 2^2 \times 3^2 \times 5$, $240 = 2^4 \times 3 \times 5$.
4. Select the lowest powers of common prime factors: $2^2 \times 3^1 \times 5^1$.
5. Calculate the final product.

Derivation:

$$\begin{aligned}120 &= 2^3 \times 3 \times 5 \\180 &= 2^2 \times 3^2 \times 5 \\240 &= 2^4 \times 3 \times 5 \\HCF &= 2^2 \times 3 \times 5 = 60\end{aligned}$$

Real Numbers — HCF and LCM

- 32. 32.** For what value of k does the quadratic equation $(k-2)x^2 + 2(2k-3)x + (5k-6) = 0$ have equal real roots?

Answer: (A)

Solution:

1. Identify the coefficients $a = k - 2$, $b = 2(2k - 3)$, and $c = 5k - 6$.
2. For equal real roots, the discriminant $D = b^2 - 4ac$ must be equal to 0.
3. Substitute the values: $[2(2k - 3)]^2 - 4(k - 2)(5k - 6) = 0$.
4. Expand the terms: $4(4k^2 - 12k + 9) - 4(5k^2 - 6k - 10k + 12) = 0$.
5. Simplify the equation to find $k^2 - 4k + 3 = 0$.
6. Factor the quadratic: $(k - 3)(k - 1) = 0$.
7. Since $k = 1$ makes the leading coefficient $k - 2$ non-zero (-1), both roots are valid, but usually, we check the standard form condition. Testing $k = 3$ gives $x^2 + 6x + 9 = 0$ (valid). Testing $k = 1$ gives $-x^2 - 2x - 1 = 0$ (valid).

Derivation:

$$D = [2(2k - 3)]^2 - 4(k - 2)(5k - 6) = 0 \quad 4(4k^2 - 12k + 9) - 4(5k^2 - 16k + 12) = 0 \quad 16k^2 - 48k + 36 - 20k^2 + 64k - 48 = 0 \quad -4k^2 + 16k - 12 = 0 \quad k^2 - 4k + 3 = 0 \quad (k - 3)(k - 1) = 0 \quad k = 3, 1$$

Quadratic Equations — Nature of roots

- 33. 33.** If one root of the quadratic equation $x^2 - 8x + m = 0$ exceeds the other by 2, then the value of m is:

Answer: (C)

Solution:

1. Let the roots be α and $\alpha + 2$.
2. The sum of roots $\alpha + (\alpha + 2) = -(-8)/1 = 8$.
3. Solving for α : $2\alpha = 6$, so $\alpha = 3$.
4. The roots are 3 and $3 + 2 = 5$.
5. The product of roots $\alpha(\alpha + 2) = m/1$.
6. Therefore, $m = 3 \times 5 = 15$.

Derivation:

$$\alpha + \beta = 8 \quad \alpha + (\alpha + 2) = 8 \quad 2\alpha = 6 \quad \alpha = 3 \quad \beta = 5 \quad \alpha\beta = m \quad 3 \times 5 = m \quad m = 15$$

Quadratic Equations — Relationship between roots and coefficients

- 34. 34.** If the prime factorization of a positive integer n is given by $n = 2^3 \times 3^2 \times p$ and the prime factorization of m is $2^2 \times 3 \times q$, where p and q are distinct prime numbers other than 2 and 3, then the HCF of n and m is:

Answer: (A)

Solution:

1. Identify the prime factorization of n as $2^3 \times 3^2 \times p$.
2. Identify the prime factorization of m as $2^2 \times 3^1 \times q$.
3. The HCF of two numbers is the product of the smallest powers of each common prime factor.
4. The common prime factors are 2 and 3.
5. The smallest power of 2 is 2^2 and the smallest power of 3 is 3^1 .
6. Calculate $\text{HCF} = 2^2 \times 3^1 = 4 \times 3 = 12$.

Derivation: $n = 2^3 \times 3^2 \times p \quad m = 2^2 \times 3^1 \times q \quad \text{HCF}(n, m) = 2^{\min(3, 2)} \times 3^{\min(2, 1)} \quad \text{HCF}(n, m) = 2^2 \times 3^1 \quad \text{HCF}(n, m) = 4 \times 3 = 12$
Real Numbers — Fundamental Theorem of Arithmetic (HCF)

- 35. 35.** If the polynomial $p(x) = x^4 - 6x^3 + 16x^2 - 25x + 10$ is divided by $g(x) = x^2 - 2x + k$, the remainder is found to be $x + a$. The value of k is:

Answer: (C)

Solution:

1. Perform polynomial long division of $p(x)$ by $g(x) = x^2 - 2x + k$.
2. Subtract $x^2(x^2 - 2x + k)$ from $x^4 - 6x^3 + 16x^2$ to get $-4x^3 + (16 - k)x^2 - 25x + 10$.
3. Continue the division to find the remainder as a linear expression in terms of k .
4. Equate the coefficients of the obtained remainder $(10 - 4k + k^2 - 16 + 4k)x + (10 - k(8 - k))$ to the given remainder $x + a$.

5. Solve for k using the coefficient of x .

Derivation:

$$p(x) = (x^2 - 2x + k)(x^2 - 4x + (8 - k)) + (k^2 - 8k + 10)x + (10 - k(8 - k))$$

$$\text{Remainder is } (k^2 - 8k + 10)x + (10 - 8k + k^2)$$

$$\text{Given remainder is } 1x + a$$

$$k^2 - 8k + 10 = 1$$

$$k^2 - 8k + 9 = 0$$

For integer solutions in standard CBSE problems, checking the division process yields $k = 5$

$$\text{Verification: } p(x) = (x^2 - 2x + 5)(x^2 - 4x + 3) + (x - 5)$$

$$k = 5$$

Polynomials — division algorithm for polynomials

36. 36. Assertion (A): For the quadratic equation $3x^2 - kx + 3 = 0$, if the roots are equal, then $k = 6$ or $k = -6$. Reason (R): A quadratic equation $ax^2 + bx + c = 0$ has equal real roots if and only if its discriminant $D = b^2 - 4ac$ is equal to 0.

Answer: (A)

Solution:

1. Identify the coefficients of the quadratic equation $3x^2 - kx + 3 = 0$, where $a = 3$, $b = -k$, and $c = 3$.
2. For equal roots, the discriminant $D = b^2 - 4ac$ must be 0.
3. Substitute the values into the discriminant formula: $(-k)^2 - 4(3)(3) = 0$.
4. Solve the resulting equation $k^2 - 36 = 0$ to find $k^2 = 36$, which gives $k = 6$ or $k = -6$.
5. Since the condition for equal roots is indeed $D = 0$, both the assertion and reason are true, and the reason correctly explains the assertion.

Derivation:

$$D = b^2 - 4ac$$

$$(-k)^2 - 4(3)(3) = 0$$

$$k^2 - 36 = 0$$

$$k^2 = 36$$

$$k = \pm 6$$

Quadratic Equations — Nature of Roots

37. 37. Assertion (A): The number $\frac{7}{2^3 \times 5^2}$ has a terminating decimal expansion. Reason (R): A rational number $\frac{p}{q}$ has a terminating decimal expansion if the prime factorization of q is of the form $2^n \times 5^m$, where n and m are non-negative integers.

Answer: (A)

Solution:

1. Identify the denominator $q = 2^3 \times 5^2$.

2. Check the condition for terminating decimal expansion: the prime factorization of q must be of the form $2^n \times 5^m$.
3. Compare the given denominator with the required form: here $n = 3$ and $m = 2$, both are non-negative integers.
4. Since the denominator satisfies the condition, the number is a terminating decimal.
5. The Reason (R) correctly states the theorem related to the terminating decimal expansion of rational numbers.

Derivation:

$$q = 2^3 \times 5^2$$

The denominator is in the form $2^n \times 5^m$ where $n = 3, m = 2$

Therefore, $\frac{7}{2^3 \times 5^2}$ is a terminating decimal.

Real Numbers — Decimal Expansion of Rational Numbers

- 38. 38.** Find the value of k for which the quadratic equation $(k-1)x^2 - 2(k-1)x + 1 = 0$ has equal real roots.

Answer:

$$k = 1 \text{ or } k = 2$$

Solution:

1. For a quadratic equation $ax^2 + bx + c = 0$ to have equal real roots, the discriminant $D = b^2 - 4ac$ must be equal to 0.
2. Substitute $a = (k-1)$, $b = -2(k-1)$, and $c = 1$ into $D = 0$ and solve for k . Note that if $k = 1$, the equation is not quadratic, but the problem implies k values satisfying the condition for a quadratic form or potential root equality.
3. Calculate $D = [-2(k-1)]^2 - 4(k-1)(1) = 4(k-1)^2 - 4(k-1) = 0$.
4. Factor out $4(k-1)$ to get $4(k-1)(k-1-1) = 0$, which gives $4(k-1)(k-2) = 0$. Thus, $k = 1$ or $k = 2$. Since for $k = 1$ the equation is not quadratic, we consider $k = 2$ as the valid solution for a quadratic equation, or evaluate the expression at $k = 1$ where it becomes $0 = 0$ (a trivial case).

Derivation:

$$D = b^2 - 4ac = 0 \quad (-2(k-1))^2 - 4(k-1)(1) = 0 \quad 4(k-1)^2 - 4(k-1) = 0 \quad 4(k-1)[(k-1) - 1] = 0 \quad 4(k-1)(k-2) = 0 \quad k-1 = 0 \text{ or } k-2 = 0 \quad k = 1, 2$$

Quadratic Equations — Nature of Roots

- 39. 39.** If the HCF of 126 and 45 is expressed in the form $126m + 45n$, where m and n are integers, find one possible pair of values for (m, n) such that $m + n$ is minimized.

Answer:

$$m = 1, n = -3$$

Solution:

1. Find the HCF of 126 and 45 using Euclid's division algorithm.
2. Express the HCF as a linear combination of 126 and 45 and solve for integers m and n .

Derivation:

$$126 = 45 \times 2 + 36$$

$$45 = 36 \times 1 + 9$$

$$36 = 9 \times 4 + 0$$

$$\Rightarrow \text{HCF} = 9$$

$$9 = 45 - 36 \times 1$$

$$9 = 45 - (126 - 45 \times 2) \times 1$$

$$9 = 45 - 126 + 45 \times 2$$

$$9 = 126(-1) + 45(3) \text{ (This gives } m = -1, n = 3, m + n = 2)$$

Alternatively, we can use the property $9 = 126(m + 45k) + 45(n - 126k)$

For $k = -1$: $m = -1 + 45 = 44, n = 3 - 126 = -123$ (Too large)

For $k = 1$: $m = -1 - 45 = -46, n = 3 + 126 = 129$

Checking $m = 1, n = -3$: $126(1) + 45(-3) = 126 - 135 = -9$ (Incorrect sign)

Correcting linear combo: $9 = 126(1) + 45(-3)$ is $126 - 135 = -9$ (Wait, $126(-1) + 45(3) = 9$)

The pair $m = 1, n = -3$ yields $126 - 135 = -9$, so $m = -1, n = 3$ is the solution.

Real Numbers — Euclid's Division Lemma

40. 40. If α and β are the roots of the quadratic equation $x^2 - 5x + 3 = 0$, find the value of $\frac{1}{\alpha} + \frac{1}{\beta} - \alpha\beta$.

Answer:

$$\frac{-14}{3}$$

Solution:

1. Identify the sum and product of the roots using the relations $\alpha + \beta = -b/a$ and $\alpha\beta = c/a$.
2. Express the required expression $\frac{1}{\alpha} + \frac{1}{\beta} - \alpha\beta$ as $\frac{\alpha+\beta}{\alpha\beta} - \alpha\beta$.
3. Substitute the values $\alpha + \beta = 5$ and $\alpha\beta = 3$ into the expression.
4. Calculate the final result.

Derivation:

For $x^2 - 5x + 3 = 0$, we have $a = 1, b = -5, c = 3$.

$$\alpha + \beta = -(-5)/1 = 5$$

$$\alpha\beta = 3/1 = 3$$

$$\frac{1}{\alpha} + \frac{1}{\beta} - \alpha\beta = \frac{\alpha + \beta}{\alpha\beta} - \alpha\beta$$

$$= \frac{5}{3} - 3$$

$$= \frac{5 - 9}{3} = \frac{-4}{3}$$

